

Bayesian Inference: Modeling and Computation

This chapter explores several practical issues for a Bayesian approach to inference. The first section explores an approach used to specify prior distributions called *hierarchical modeling*, based on hyperparameters and conditioning. Section 15.2 discusses the robustness to the choice of prior distribution. Sections 15.4 and 15.5 deal with the Metropolis–Hastings algorithm and the Gibbs sampler, simulation methods that can be used to approximate posterior expectations numerically. As background, Section 15.3 provides a brief introduction to Markov chains. Finally, Section 15.6 illustrates how Gibbs sampling can be used in a Bayesian approach to image processing.

15.1 Hierarchical Models

Hierarchical modeling is a mixture approach to setting a prior distribution in stages. It arises when there is a natural family of prior distributions $\{\Lambda_\tau : \tau \in \mathbb{T}\}$ for our unknown parameter Θ . If the value τ characterizing the prior distribution is viewed as an unknown parameter, then for a proper Bayesian analysis τ should be viewed as a realization of an unknown random variable $\mathcal{T}$. With this approach, there are two random parameters, $\mathcal{T}$ and Θ . Because the distribution for our data X depends only on Θ , $\mathcal{T}$ is called a *hyperparameter*. If G is the prior distribution for $\mathcal{T}$, then the Bayesian model is completely specified by

$$\mathcal{T} \sim G, \quad \Theta | \mathcal{T} = \tau \sim \Lambda_\tau,$$

and

$$X | \mathcal{T} = \tau, \Theta = \theta \sim P_\theta.$$

Note that in this model,

$$P(\Theta \in B) = EP(\Theta \in B | \mathcal{T}) = E\Lambda_{\mathcal{T}}(B) = \int \Lambda_\tau(B) dG(\tau),$$

which shows that the prior distribution for Θ is now a mixture of distributions in the family $\{\Lambda_\tau : \tau \in \mathbb{T}\}$. Using this mixture, the hyperparameter $\mathcal{T}$ could be eliminated from the model, although in some situations this may be counterproductive.

Example 15.1 (Compound Estimation). As a first example, let us consider the compound estimation problem considered from an empirical Bayes perspective in Section 11.1. In that section, the parameters $\Theta_1, \dots, \Theta_n$ were i.i.d. from $N(0, \tau^2)$, with the hyperparameter τ viewed as a constant. For a hierarchical Bayesian analysis, a prior distribution G would be specified for $\mathcal{T}$. Then

$$\Theta|\mathcal{T} = \tau = N(0, \tau^2 I) \text{ and } X|\mathcal{T} = \tau, \Theta = \theta \sim N(\theta, I).$$

In this example, if we eliminated the hyperparameter $\mathcal{T}$ then the prior distribution for Θ would not be conjugate and we would not be able to take advantage of exact formulas based on that structure. If the dimension n is large, numerical calculations may be a challenge. Keeping $\mathcal{T}$, smoothing leads to some simplifications. Using (11.1) the Bayes estimator for Θ is

$$\delta = E[\Theta|X] = E[E[\Theta|X, \mathcal{T}] | X] = XE\left[\frac{\mathcal{T}^2}{1 + \mathcal{T}^2} \mid X\right].$$

As noted in Section 11.1, given $\mathcal{T} = \tau$, $X_1, \dots, X_n$ are i.i.d. from $N(0, 1 + \tau^2)$, so the likelihood for τ has a simple form and compound inference can be accomplished using standard conditioning formulas to compute $E[\mathcal{T}^2/(1 + \mathcal{T}^2) | X]$. Note that all of the integrals involved are one-dimensional; therefore a numerical approach is quite feasible.

Example 15.2 (General Linear Model). The general linear model was introduced in Chapter 14. Here we consider Bayesian inference with the error variance σ^2 assumed known. This leaves β as the sole unknown parameter, viewed as a random vector $\mathcal{B}$, and

$$Y|\mathcal{B} = \beta \sim N(X\beta, \sigma^2 I),$$

with X a known $n \times p$ matrix. For a prior distribution, proceeding as in the last example we might take

$$\mathcal{B} \sim N(0, \tau^2 I).$$

If κ is the variance ratio σ^2/τ^2 , the posterior distribution is

$$\mathcal{B}|Y = y \sim N((X'X + \kappa I)^{-1}X'Y, \sigma^2(X'X + \kappa I)^{-1}).$$

As in the previous example, the posterior mean here still shrinks the UMVU $(X'X)^{-1}X'Y$ towards the origin, although the shrinkage “factor” is now the matrix $(X'X + \kappa I)^{-1}X'X$ instead of a constant.

The Bayes estimator $(X'X + \kappa I)^{-1}X'Y$ is also called a *ridge regression estimator*, originally suggested for numerical reasons to help regularize $X'X$.

As in the first example, the prior distributions $N(0, \tau^2 I)$ for $\mathcal{B}$ here are indexed by a hyperparameter τ . A hierarchical approach would model τ as a random variable $\mathcal{T}$ from some distribution G .

Example 15.3 (Testing). Another class of examples arises if natural prior distributions for θ seem appropriate under competing scientific hypotheses. In these examples the hyperparameter $\mathcal{T}$ can be a discrete variable indexing the competing theories. As a concrete example, the standard model for one-way ANOVA has $Y_{ij} \sim N(\theta_i, \sigma^2)$, $i = 1, \dots, I$, $j = 1, \dots, n_i$, with all $n = n_1 + \dots + n_I$ observations independent. If there is reason to believe that all the θ_i may be equal, then a prior distribution in which $\Theta_1 \sim N(\mu_\theta, \sigma_\theta^2)$ with all other Θ_i equal to Θ_1 may be reasonable, so

$$\Theta \sim N(\mu_\theta \mathbf{1}, \sigma_\theta^2 \mathbf{1}\mathbf{1}').$$

If instead the means differ, the prior

$$\Theta \sim N(\mu_\theta \mathbf{1}, \sigma_\theta^2 I)$$

may be more natural. If $\mathcal{T} = 1$ or 2 indexes these possibilities, then the mixture prior for Θ in a hierarchical model would be the convex combination

$$\Theta \sim P(\mathcal{T} = 1)N(\mu_\theta \mathbf{1}, \sigma_\theta^2 \mathbf{1}\mathbf{1}') + P(\mathcal{T} = 2)N(\mu_\theta \mathbf{1}, \sigma_\theta^2 I).$$

15.2 Bayesian Robustness

Ideally, in a Bayesian analysis the prior distribution is chosen to reflect a researcher's knowledge and beliefs about the unknown parameter Θ . But in practice the choice is often dictated to some degree by convenience. Conjugate priors are particularly appealing here due to simple formulas for the posterior mean. Unfortunately, the convenience of such priors entails some risk.

To explore robustness issues related to the choice of the prior distribution in a very simple setting, consider a measurement error model with

$$X|\Theta = \theta \sim N(\theta, 1).$$

Suppose the true prior distribution Λ is a t -distribution on three degrees of freedom with density

$$\lambda(\theta) = \frac{2}{\sqrt{3}\pi(1 + \theta^2/3)^2}.$$

Calculations with this prior distribution are a challenge, so it is tempting to use a conjugate normal distribution instead. The normal distribution $\Lambda_N =$

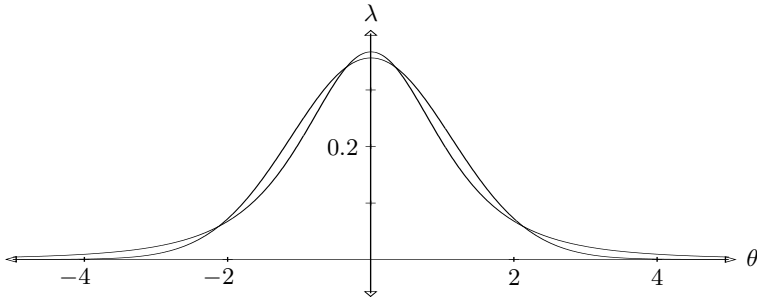


Fig. 15.1. Prior densities λ and λ_N .

$N(0, 5/4)$ seems close to Λ . Densities λ and λ_N for Λ and Λ_N , graphed in Figure 15.1, are quite similar; the largest difference between them is

$$\sup_{\theta} |\lambda(\theta) - \lambda_N(\theta)| = 0.0333,$$

and $|\Lambda(B) - \Lambda_N(B)| \leq 7.1\%$ for any Borel set B .

With squared error loss, the Bayes estimator with Λ_N as the prior is

$$\delta_N(X) = \frac{5X}{9},$$

and its risk function is

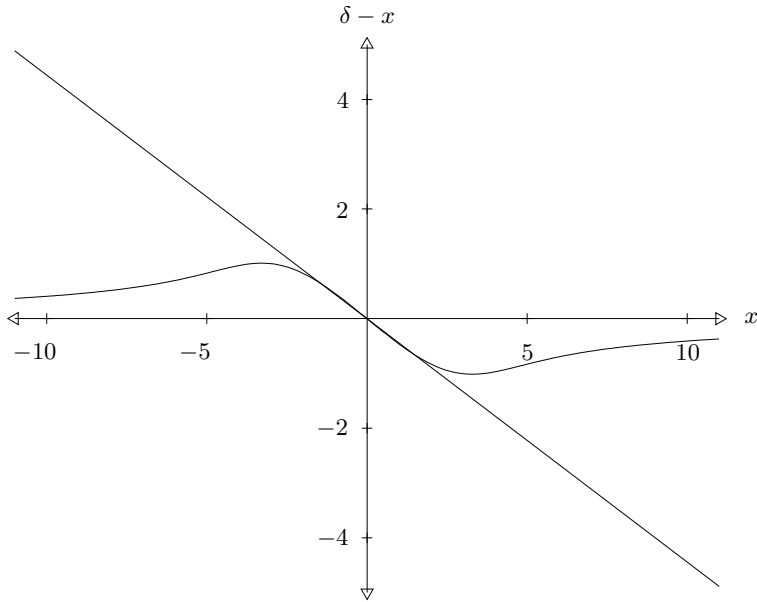


Fig. 15.2. Differences $\delta(x) - x$ and $\delta_N(x) - x$.

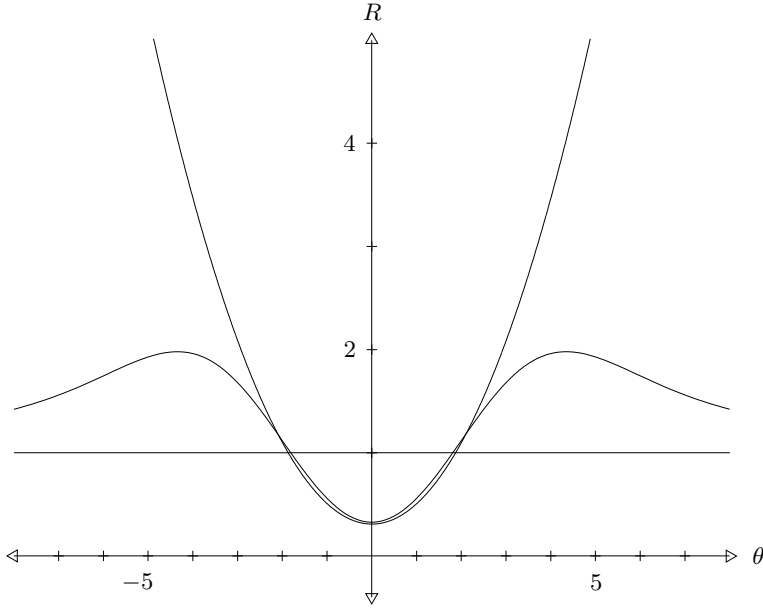


Fig. 15.3. Risk functions $R(\theta, \delta_U)$, $R(\theta, \delta_N)$, and $R(\theta, \delta)$.

$$R(\theta, \delta_N) = E[(\delta_N(X) - \theta)^2 \mid \Theta = \theta] = \frac{25 + 16\theta^2}{81}.$$

The integrated risk of δ_N under Λ_N is

$$\int R(\theta, \delta_N) d\Lambda_N(\theta) = \frac{45}{81},$$

much better than the integrated risk of 1 for the UMVU $\delta_U(X) = X$. This improvement is achieved by the shrinkage towards zero, which improves the variance of the estimator and introduces little bias when θ is near zero.

The true prior Λ has heavier tails, placing more weight on the region where δ_N is more heavily biased and its risk $R(\theta, \delta_N)$ is large. As one might guess, the Bayes estimator works to minimize risk for large θ with less shrinkage when $|X|$ is large. This can be seen in [Figure 15.2](#), which graphs the differences $\delta - \delta_U$ and $\delta_N - \delta_U$. The estimators $\delta(X)$ and $\delta_N(X)$ are very similar if $|X| < 2$, but as $|X|$ increases, $\delta(X)$ moves closer to X . In fact,

$$\delta(x) = x - 4/x + o(1/x) \quad (15.1)$$

as $x \rightarrow \pm\infty$. Thus as $\theta \rightarrow \pm\infty$, the bias of δ tends to 0 and its risk function approaches 1, the risk of δ_U , instead of increasing without bound as the quadratic risk of δ_N . [Figure 15.3](#) shows risk functions for δ_U , δ_N , and δ .

With the true prior Λ , the integrated risk for δ_N is

$$\int R(\theta, \delta_N) d\Lambda(\theta) = \frac{73}{81} = 0.901,$$

almost as high as the risk of X . The best possible integrated risk with the true prior, achieved using the Bayes estimator δ , is

$$\int R(\theta, \delta) d\Lambda(\theta) = 0.482,$$

which is 46.5% smaller than the integrated risk for δ_N .

15.3 Markov Chains

Definition 15.4. A sequence of random vectors $X_0, X_1, X_2, \dots$ taking values in a state space $(\mathcal{X}, \mathcal{A})$ form a (time homogeneous) Markov chain with transition kernel¹ Q , if

$$P(X_{n+1} \in B | X_0 = x_0, \dots, X_n = x_n) = P(X_{n+1} \in B | X_n = x_n) = Q_{x_n}(B),$$

for all $n \geq 1$, all $B \in \mathcal{A}$, and almost all $x_1, \dots, x_n$.

Using smoothing, the joint distribution of the vectors in a Markov chain can be found from the initial distribution for X_0 and the transition kernel Q . The algebra involved can be easily described introducing some convenient notation. For a probability measure π on $(\mathcal{X}, \mathcal{A})$, define a probability measure πQ by

$$\pi Q(B) = \int Q_x(B) d\pi(x). \quad (15.2)$$

Note that if π_n denotes the distribution for X_n , then by smoothing,

$$\begin{aligned} \pi_{n+1}(B) &= P(X_{n+1} \in B) = EP(X_{n+1} \in B | X_n) \\ &= EQ_{X_n}(B) = \int Q_x(B) d\pi_n(x) = \pi_n Q(B), \end{aligned}$$

and so

$$\pi_{n+1} = \pi_n Q. \quad (15.3)$$

A distribution π is called *stationary* if $\pi = \pi Q$. Using (15.3), if the initial distribution π_0 for X_0 is stationary, then $\pi_2 = \pi_1 Q = \pi_1$. Further iteration shows that $\pi_1 = \pi_2 = \pi_3 = \dots$, so in this case the random vectors in the chain are identically distributed.

If Q and $\tilde{Q}$ are transition kernels on $\mathcal{X}$, define the product kernel $Q\tilde{Q}$ by

¹ The kernel Q should satisfy the usual regularity conditions for stochastic transition kernels: Q_x should be a probability measure on $(\mathcal{X}, \mathcal{A})$ for all $x \in \mathcal{X}$, and $Q_x(B)$ should be a measurable function of x for all $B \in \mathcal{A}$.

$$(Q\tilde{Q})_x = Q_x\tilde{Q}. \quad (15.4)$$

Taking $Q^2 \stackrel{\text{def}}{=} QQ$, by smoothing,

$$\begin{aligned} P(X_{n+2} \in B | X_0 = x_0, \dots, X_n = x_n) \\ = \int P(X_{n+2} \in B | X_0 = x_0, \dots, X_{n+1} = x_{n+1}) dQ_{x_n}(x_{n+1}) \\ = Q_{x_n}^2(B), \end{aligned}$$

and so Q^2 can be viewed as the two-step transition kernel for the Markov chain. Similarly, the k -fold product Q^k gives chances for k -step transitions:

$$Q_{x_n}^k(B) = P(X_{n+k} \in B | X_0 = x_0, \dots, X_n = x_n).$$

Example 15.5. If the state space $\mathcal{X}$ is finite, $\mathcal{X} = \{1, \dots, m\}$ say, then a distribution π on $\mathcal{X}$ can be specified through its mass function given as a row vector

$$\pi = [\pi(\{1\}), \dots, \pi(\{m\})],$$

and the transition kernel Q can be specified by a matrix $\mathbf{Q}$ with

$$\mathbf{Q}_{ij} = Q_i(\{j\}) = P(X_{n+1} = j | X_n = i).$$

If we let π_n be the row vector for the distribution of X_n , then

$$\begin{aligned} [\pi_{n+1}]_j &= P(X_{n+1} = j) = EP(X_{n+1} = j | X_n) = EQ_{X_n}(\{j\}) \\ &= \sum_i P(X_n = i)Q_i(\{j\}) = \sum_i [\pi_n]_i \mathbf{Q}_{ij} = [\pi_n \mathbf{Q}]_j. \end{aligned}$$

So

$$\pi_{n+1} = \pi_n \mathbf{Q}.$$

Thus, if distributions are represented as row vectors and transition kernels are represented as matrices, the “multiplication” in (15.3) becomes ordinary matrix multiplication. Similarly, if $\mathbf{Q}$ and $\tilde{\mathbf{Q}}$ are matrices corresponding to transition kernels Q and $\tilde{Q}$, then

$$(Q\tilde{Q})_i(\{j\}) = \int \tilde{Q}_x(\{j\}) dQ_i(x) = \sum_x \mathbf{Q}_{ix} \tilde{\mathbf{Q}}_{xj} = [\mathbf{Q}\tilde{\mathbf{Q}}]_{ij}$$

and the matrix representing $Q\tilde{Q}$ is simply the matrix product $\mathbf{Q}\tilde{\mathbf{Q}}$.

For a finite Markov chain, because the mass function for Q_i sums to 1, we have

$$\sum_j \mathbf{Q}_{ij} = [\mathbf{Q}\mathbf{1}]_i = 1,$$

and hence

$$\mathbf{Q}\mathbf{1} = \mathbf{1}.$$

This shows that $\mathbf{1}$ is a right-eigenvector for $\mathbf{Q}$ with unit eigenvalue. Since $\pi\mathbf{Q}$ is given by matrix multiplication, a probability distribution π will be stationary if

$$\pi = \pi\mathbf{Q},$$

that is, if π is a left-eigenvector of $\mathbf{Q}$ with unit eigenvalue. In general, if λ is an eigenvalue for $\mathbf{Q}$, then $|\lambda| \leq 1$,

Convergence properties for finite Markov chains are commonly related to the Frobenius theory for positive matrices, covered in Appendix 2 of Karlin and Taylor (1975). If A is an $n \times n$ matrix with eigenvalues $\lambda_1, \dots, \lambda_n$, then its *spectral radius* is defined as $r = \max\{|\lambda_1|, \dots, |\lambda_n|\}$.

Theorem 15.6 (Perron–Frobenius). *Let A be an $n \times n$ matrix with non-negative entries and spectral radius r , and assume that $A^m > 0$ for some $m > 0$. Then*

1. *The spectral radius r is a simple² eigenvalue for A , $r > 0$, and if λ is any other eigenvalue, $|\lambda| < r$.*
2. *There are left- and right-eigenvectors associated with r with positive entries. Specifically, there is a row vector v and a column vector w with $v > 0$, $w > 0$, $vA = rv$, and $Aw = rw$.*
3. *If v is normalized so that its entries sum to 1 and w is normalized so that $vw = 1$, then*

$$r^{-n}A^n \rightarrow vw$$

as $n \rightarrow \infty$.

4. *The spectral radius r satisfies*

$$\min_i \sum_j A_{ij} \leq r \leq \max_i \sum_j A_{ij}.$$

To characterize eigenvalues and convergence properties for finite chains in regular cases, we need a few definitions. Let

$$L_x(A) = P(X_n \in A, \exists n \geq 1 | X_0 = x),$$

the chance the chain ever visits A if it starts at x . States i and j for a finite chain are said to *communicate* if the chain can move from either of the states to the other; that is, if $L_i(\{j\}) > 0$ and $L_j(\{i\}) > 0$. If all of the states communicate, the chain is called *irreducible*. The chain is called *periodic* if $\mathcal{X}$ can be partitioned into sets $\mathcal{X}_1, \dots, \mathcal{X}_k$, $k \geq 2$, and the process cycles between these sets: if $i \in \mathcal{X}_j$, $1 \leq j \leq k-1$,

$$Q_i(\mathcal{X}_{j+1}) = P(X_1 \in \mathcal{X}_{j+1} | X_0 = i) = 1,$$

and if $i \in \mathcal{X}_k$, $Q_i(\mathcal{X}_1) = 1$.

² An eigenvalue is simple if it is a simple root of the characteristic equation. In this case, eigenspaces (left or right) will be one-dimensional.

For finite chains, properties needed for simulation arise when the chain is irreducible and aperiodic. In this case, it is not hard to argue that $\mathbf{Q}^m > 0$, so $\mathbf{Q}$ satisfies the conditions of the Perron–Frobenius theorem. Because $\mathbf{Q}\mathbf{1} = \mathbf{1}$, the spectral radius r for $\mathbf{Q}$ must be $r = 1$ by the fourth assertion of the theorem. Because r is a simple eigenvalue, there will exist a unique corresponding left-eigenvector $\boldsymbol{\pi}$ with entries summing to 1, corresponding to a unique stationary distribution π for the chain. The mass function for this distribution can be found by solving the linear equations

$$\boldsymbol{\pi} = \boldsymbol{\pi}\mathbf{Q} \text{ and } \boldsymbol{\pi}\mathbf{1} = 1.$$

By the third assertion in the theorem,

$$\mathbf{Q}^n \rightarrow \mathbf{1}\boldsymbol{\pi}$$

as $n \rightarrow \infty$. In probabilistic terms, this means that

$$X_n \Rightarrow \pi$$

as $n \rightarrow \infty$, regardless of the initial state (or distribution) for the chain. The stationary distribution also gives the long run proportion of time the process spends in the various states, and the following law of large numbers holds:

$$\frac{1}{n} \sum_{i=1}^n f(X_i) \rightarrow \int f d\pi,$$

with probability one, regardless of the initial distribution π_0 . Using this, the value for $\int f d\pi$ can be approximated by simulation, having a computer generate the chain numerically and averaging the values for f . For an extended discussion of finite chains, see Kemeny and Snell (1976).

The theory for Markov chains when $\mathcal{X}$ is infinite but denumerable is similar, although irreducible and aperiodic chains without a stationary distribution are possible; see Karlin and Taylor (1975) or another introduction to stochastic processes. When $\mathcal{X}$ is not denumerable, the relevant theory, presented in Nummelin (1984) or Meyn and Tweedie (1993), is much more complicated. For simulation, the most appealing notion of regularity might be Harris recurrence. Tierney (1994) gives convergence results for the Metropolis–Hastings algorithm and the Gibbs sampler, discussed in the next two sections.

15.4 Metropolis–Hastings Algorithm

In a Bayesian model, if θ has density λ and the conditional density of X given $\theta = \theta$ is p_θ , then the posterior density of θ given $X = x$ is proportional (in θ) to

$$\lambda(\theta)p_\theta(x).$$

To compute the posterior density $\lambda(\theta|x)$ we should divide this function by its integral,

$$\int \lambda(\theta)p_\theta(x) d\theta.$$

In practice, this integral may be difficult to evaluate, explicitly or numerically, especially if Θ is multidimensional. The Metropolis–Hastings algorithm is a simulation method that allows approximate sampling from this posterior distribution without computing the normalizing constant. Specifically, the algorithm gives a Markov chain that has the target law as its stationary distribution.

To describe the transition kernel Q for the Markov chain, let π denote a target distribution on some state space $\mathcal{X}$ with density f with respect to a dominating measure μ . The chain runs by accepting or rejecting potential states generated using a transition kernel J with densities $j_x = dJ_x/d\mu$. The chances for accepting or rejecting a new value are based on a function r given by

$$r(x_0, x^*) = \frac{f(x^*)/j_{x_0}(x^*)}{f(x_0)/j_{x^*}(x_0)}.$$

Note that r can be computed if f is only known up to a proportionality constant. Let X_0 denote the initial state of the chain. Given $X_0 = x_0$, a variable X^* is drawn from J_{x_0} , so

$$X^*|X_0 = x_0 \sim J_{x_0}.$$

The next state for the Markov chain, X_1 , will be either X_0 or X^* , with

$$P(X_1 = X^*|X_0 = x_0, X^* = x^*) = \min\{r(x_0, x^*), 1\}.$$

Thus

$$\begin{aligned} P(X_1 \in A|X_0 = x_0, X^* = x^*) \\ = 1_A(x^*) \min\{r(x_0, x^*), 1\} + 1_A(x_0)(1 - \min\{r(x_0, x^*), 1\}). \end{aligned}$$

Integrating against the conditional distribution for X^* given $X_0 = x_0$, by smoothing

$$\begin{aligned} Q_{x_0}(A) &\stackrel{\text{def}}{=} P(X_1 \in A|X_0 = x_0) \\ &= 1_A(x_0) + \int_A \min\{r(x_0, x^*), 1\} j_{x_0}(x^*) d\mu(x^*) \\ &\quad - 1_A(x_0) \int \min\{r(x_0, x^*), 1\} j_{x_0}(x^*) d\mu(x^*). \end{aligned}$$

To check that π is a stationary distribution for the chain with transition kernel Q we need to show that if $X_0 \sim \pi$, then $X_1 \sim \pi$. If $X_0 \sim \pi$, then by smoothing,

$$P(X_1 \in A) = EP(X_1 \in A|X_0) = EQ_{X_0}(A) = \int Q_{x_0}(A) d\pi(x_0).$$

Because $\int 1_A(x_0) d\pi(x_0) = \pi(A)$, this will hold if

$$\begin{aligned} & \iint \min\{r(x_0, x^*), 1\} 1_A(x^*) f(x_0) j_{x_0}(x^*) d\mu(x_0) d\mu(x^*) \\ &= \iint \min\{r(x_0, x^*), 1\} 1_A(x_0) f(x_0) j_{x_0}(x^*) d\mu(x_0) d\mu(x^*). \end{aligned}$$

Using the formula for r , this equation becomes

$$\begin{aligned} & \iint \min\{f(x^*) j_{x^*}(x_0), f(x_0) j_{x_0}(x^*)\} 1_A(x^*) d\mu(x_0) d\mu(x^*) \\ &= \iint \min\{f(x^*) j_{x^*}(x_0), f(x_0) j_{x_0}(x^*)\} 1_A(x_0) d\mu(x_0) d\mu(x^*), \end{aligned}$$

which holds by Fubini's theorem.

Convergence of the Metropolis–Hastings algorithm is discussed in Tierney (1994). Turning to practical considerations, several things should be considered in choosing the jump kernel J . First, it should be easy to sample values from J_x , and the formula to compute r should be simple. In addition, J should move easily to all relevant areas of the state space and jumps should not be rejected too often.

15.5 Gibbs Sampler

The Gibbs sampler is based on alternate sampling from the conditional distributions for the target distribution π . If $(X, Y) \sim \pi$, let R denote the conditional distribution of X given Y , and let $\tilde{R}$ denote the conditional distribution of Y given X . If (X_0, Y_0) is the initial state for the Markov chain, then we find (X_1, Y_1) by first sampling X_1 from R and then drawing Y_1 from $\tilde{R}$. Specifically,

$$X_1|X_0 = x_0, Y_0 = y_0 \sim R_{y_0}$$

and

$$Y_1|X_0 = x_0, Y_0 = y_0, X_1 = x_1 \sim \tilde{R}_{x_1}.$$

Continuing in this fashion, (X_i, Y_i) , $i \geq 0$ is a Markov chain.

The Gibbs sampler can be easily extended to joint distributions for more than two variables (or vectors). If we are interested in simulating the joint distribution of X , Y , and Z , say, we could generate a new X sampling from the conditional distribution for X given Y and Z , then generate a new Y from the conditional distribution of Y given X and Z , then generate a new Z from the conditional distribution of Z given X and Y , and so on.

The Gibbs sampler is useful in simulation mainly through dimension reduction. Sampling from a univariate distribution is typically much easier than multivariate sampling. If a better approach is not available, univariate simulation is possible using the probability integral transformation whenever the cumulative distribution function is available. Also, note that if the target distribution π is absolutely continuous with density f proportional to a known function g , then, in order to compute f from g , we would normalize g dividing by its integral,

$$\iint g(x, y) dx dy.$$

In contrast, to find conditional densities we would normalize g dividing by

$$\int g(x, y) dx \text{ or } \int g(x, y) dy.$$

The normalization for the conditional distributions needed for Gibbs sampling involves univariate integration instead of the multiple integration needed to find the joint density.

To check that the Gibbs sampler has π as a stationary distribution, let π_X and π_Y denote the marginal distributions of X and Y when $(X, Y) \sim \pi$. By smoothing,

$$\begin{aligned} P[(X, Y) \in A] &= EE[1_A(X, Y) \mid Y] \\ &= \iint 1_A(x, y) dR_y(x) d\pi_Y(y), \end{aligned} \quad (15.5)$$

and reversing X and Y ,

$$P[(X, Y) \in A] = \iint 1_A(x, y) d\tilde{R}_x(y) d\pi_X(x). \quad (15.6)$$

Suppose we start the chain with distribution π , so $(X_0, Y_0) \sim (X, Y)$. Then by smoothing, since $Y_0 \sim \pi_Y$ and the conditional distribution of X_1 given Y_0 is R ,

$$P[(X_1, Y_0) \in A] = EE[1_A(X_1, Y_0) \mid Y_0] = \iint 1_A(x, y) dR_y(x) d\pi_Y(y).$$

Comparing this with (15.5), $(X_1, Y_0) \sim \pi$. In particular, $X_1 \sim \pi_X$. Smoothing again, since $\tilde{R}$ is the conditional distribution of Y_1 given X_1 ,

$$\begin{aligned} P[(X_1, Y_1) \in A] &= EE[1_A(X_1, Y_1) \mid X_1] \\ &= \iint 1_A(x, y) d\tilde{R}_x(y) d\pi_X(x). \end{aligned}$$

Comparing this with (15.6), $(X_1, Y_1) \sim (X, Y) \sim \pi$, which shows that π is stationary.

15.6 Image Restoration

Gibbs sampling was introduced in a landmark paper by Geman and Geman (1984) on Bayesian image restoration. The example here is based on this work with a particularly simple form for the prior distribution. The unknown image is represented by unknown greyscale values Θ_z at nm pixels $z = (i, j)$ in a rectangular grid $\mathcal{T}$:

$$z = (i, j) \in \mathcal{T} \stackrel{\text{def}}{=} \{1, \dots, m\} \times \{1, \dots, n\}.$$

Given $\Theta = \theta$, observed data X_z , $z \in \mathcal{T}$, are independent with

$$X_z \sim N(\theta_z, \sigma^2), \quad z \in \mathcal{T}, \quad (15.7)$$

and σ^2 considered known.

In real images, greyscale values at nearby pixels are generally highly correlated, whereas well-separated pixels are nearly uncorrelated. For good performance, correlations for the prior distribution for Θ should have similar form. For simplicity we restrict attention here to normal distributions. In one dimension, the autoregressive model in Example 6.4 has these features; it is not hard to show that $\text{Cor}(X_i, X_j) = \rho^{|i-j|}$. The joint density in that example is proportional to

$$\exp \left[-\frac{a}{2} \sum_{i=1}^n x_i^2 + b \sum_{i=1}^{n-1} x_i x_{i+1} \right],$$

where the constants a and b satisfy $|b| < a/2$, which ensures that this expression is integrable. To construct a prior density for Θ with a similar form, call a set of two pixels $\{z_1, z_2\} \in \mathcal{T}^2$ an *edge* if $\|z_1 - z_2\| = 1$ and let $\mathcal{E}$ denote the set of all edges. The priors of interest here have form

$$\lambda(\theta) \propto_{\theta} \exp \left[-\frac{a}{2} \sum_z \theta_z^2 + b \sum_{\{z_1, z_2\} \in \mathcal{E}} \theta_{z_1} \theta_{z_2} \right]. \quad (15.8)$$

For integrability, assume $|b| < a/4$. Neglecting effects that arise near the edge of the image,

$$\sigma_{\Theta}^2 \stackrel{\text{def}}{=} \text{Var}(\Theta_z) = \frac{2K(\sqrt{\eta})}{a\pi},$$

where $\eta = 4b/a$ and K is the complete elliptic integral of the first kind, given³ by

$$K(x) = \int_0^{\pi/2} \frac{d\phi}{\sqrt{1 - x^2 \sin^2(\phi)}} = \frac{\pi}{2} \sum_{n=0}^{\infty} \left(\frac{x^2}{16} \right)^n \binom{2n}{n}^2.$$

Also, if z_1 and z_2 are adjacent pixels, then

³ Different sources use slightly different definitions.

$$\rho \stackrel{\text{def}}{=} \text{Cor}(\Theta_{z_1}, \Theta_{z_2}) = \frac{2K(\sqrt{\eta}) - \pi}{2\eta K(\sqrt{\eta})}.$$

Solving,

$$2K(\sqrt{\eta}) = \frac{\pi}{1 - \rho\eta}.$$

As η increases to 1, $K(\sqrt{\eta})$ increases without bound, and so $\rho \uparrow 1$ as $\eta \uparrow 1$. By results in Cody (1965), $K(\sqrt{\eta}) \sim \log[4/(1 - \eta)]$ as $\eta \uparrow 1$. From this, when ρ is near 1,

$$\eta \approx 1 - 4 \exp\left[-\frac{\pi}{2(1 - \rho)}\right],$$

and so

$$a \approx \frac{1}{\sigma_\Theta^2(1 - \rho)} \quad \text{and} \quad b \approx \frac{a}{4} - a \exp\left[-\frac{\pi}{2(1 - \rho)}\right], \quad (15.9)$$

relating a and b in the prior density to σ_Θ^2 and ρ .

The prior density in (15.8) has an interesting and useful structure. Suppose we let $z_1, \dots, z_4$ denote the pixels adjacent to some pixel z . If we fix the values for θ at these pixels, then λ factors into a function of θ_z and a function of θ s at the remaining $nm - 5$ pixels. From this, given $\Theta_{z_1}, \dots, \Theta_{z_4}$, Θ_z is conditionally independent of the image values at the remaining pixels. This conditional independence might be considered a Markov property, and the distribution for Θ here is called a *Markov random field*. Building on this idea, let us divide $\mathcal{T}$ into “even” and “odd” pixels:

$$\mathcal{T}_1 = \{(i, j) \in \mathcal{T} : i + j \text{ odd}\} \quad \text{and} \quad \mathcal{T}_2 = \{(i, j) \in \mathcal{T} : i + j \text{ even}\}.$$

If we fix the values for θ_z , $z \in \mathcal{T}_1$, then $\lambda(\theta)$ has form $\prod_{z \in \mathcal{T}_2} f_z(\theta_z)$. Thus given Θ_z , $z \in \mathcal{T}_1$, the Θ_z , $z \in \mathcal{T}_2$, are conditionally independent. Below we show that posterior distributions have this same structure.

Taking $\tau = 1/\sigma^2$, the “precision” of the X_z , by (15.7) the density for X given $\Theta = \theta$ is

$$p_\theta(x) \propto_\theta \exp\left[-\frac{\tau}{2} \sum_{z \in \mathcal{T}} \theta_z^2 + \tau \sum_{z \in \mathcal{T}} \theta_z x_z\right].$$

Therefore the conditional density for Θ given $X = x$ is

$$\lambda(\theta|x) = c(x) \exp\left[-\frac{a + \tau}{2} \sum_{z \in \mathcal{T}} \theta_z^2 + \tau \sum_{z \in \mathcal{T}} \theta_z x_z + b \sum_{\{z_1, z_2\} \in \mathcal{E}} \theta_{z_1} \theta_{z_2}\right], \quad (15.10)$$

with $c(x)$ chosen as usual so that $\int \cdots \int \lambda(\theta|x) d\theta = 1$. With the quadratic structure, this conditional density must be normal. The mean can be found solving linear equations to minimize the quadratic function of θ in the exponent, and the covariance is minus one half the inverse of the matrix defining

the quadratic form in θ . With modern computing these calculations may be possible, but the $nm \times nm$ matrices involved, although sparse, are very large. Thus Gibbs sampling may be an attractive alternative.

To implement Gibbs sampling, we need to determine the relevant conditional distributions. Let $\mathcal{N}_z$ denote the pixels neighboring $z \in \mathcal{T}$,

$$\mathcal{N}_z = \{\tilde{z} \in \mathcal{T} : \|\tilde{z} - z\| = 1\},$$

and define

$$S_z = \sum_{\tilde{z} \in \mathcal{N}_z} \theta_{\tilde{z}},$$

the sum of the θ values at pixels neighboring z . Isolating the terms in (15.10) that depend on θ_z , $\lambda(\theta|x)$ is

$$\exp\left[-\frac{a+\tau}{2}\theta_z^2 + \theta_z(\tau x_z + bS_z)\right] \quad (15.11)$$

times a term that is functionally independent of θ_z . As a function of θ_z , the expression in (15.11) is proportional to a normal density with variance $1/(a+\tau)$ (or precision $a+\tau$) and mean

$$\frac{\tau x_z + bS_z}{a+\tau}.$$

With the product structure, given $\Theta_{\tilde{z}} = \theta_{\tilde{z}}$, $\tilde{z} \in \mathcal{T}_1$, the Θ_z for $z \in \mathcal{T}_2$ are conditionally independent with

$$\Theta_z | \Theta_{\tilde{z}} = \theta_{\tilde{z}}, \tilde{z} \in \mathcal{T}_1, X = x \sim N\left(\frac{\tau x_z + bS_z}{a+\tau}, \frac{1}{a+\tau}\right), \quad z \in \mathcal{T}_2, \quad (15.12)$$

with a similar result for the conditional distribution of the image given values at pixels in $\mathcal{T}_2$.

The conditional distributions just described are exactly what we need to implement Gibbs sampling from the posterior distribution of the image. Starting with image values at pixels in $\mathcal{T}_1$, independent values at pixels in $\mathcal{T}_2$ would be drawn using the conditional marginal distributions in (15.12). Reversing the sets $\mathcal{T}_1$ and $\mathcal{T}_2$, values at pixels in $\mathcal{T}_1$ would next be drawn independently from the appropriate normal distributions. Iterating, the posterior mean should be close to the average values in the simulation.

To illustrate how this approach works in practice, let us consider a numerical example. The true θ is a 99×64 image of the letter A, displayed as the first image in Figure 15.4. The value for θ at “dark” pixels is 0 and the value at “light” pixels is 5. The second image in Figure 15.4 shows the raw data X , drawn from a normal distribution with mean θ and covariance $9I$, so $\sigma = 3$.

By symmetry, the mean for the prior distribution λ in (15.8) is zero. But the average grayscale value in the true image θ is 4.1477, which is significantly different. It seems natural, although a bit ad hoc, to center the raw data by



Fig. 15.4. Left to right: True image θ , raw data X , undersmoothed, matched covariance, oversmoothed.

subtracting the overall average $\bar{X} = \sum_{z \in \mathcal{T}} X_z / (nm)$ from the greyscale value at each pixel, before proceeding with our Bayesian analysis. After processing, we can add $\bar{X}$ back to the posterior mean. Results doing this should be similar to those obtained using a normal prior with mean $\bar{X}$ for Θ , or (more properly) following a hierarchical approach in which the mean for Θ is an additional hyperparameter and this parameter has a reasonably diffuse distribution.

Empirical estimates for σ_Θ^2 and ρ , based on the true θ , are 3.5350 and 0.8565, respectively. Using (15.9) these values correspond to $a = 1.971$ and $b = 0.493$. With these values, the prior variance and covariance between values at adjacent pixels will match the empirical values for the true image, and it seems reasonable to hope for excellent restoration using this prior. Of course in practice the true image and associated moments are unknown. For comparison, we have also done an analysis with two other priors. In both, we take $\sigma_\Theta^2 = 3.5350$, matching the empirical variance for the true θ . But in one of the priors we take $\rho = 0.70$ and in the other we take $\rho = 0.95$. Since higher values for ρ give smoother images Θ , we anticipate that the posterior mean will undersmooth X when $\rho = 0.70$ and oversmooth X when $\rho = 0.95$. The final three images in Figure 15.4 show posterior means for these three prior distributions, which can be found by Gibbs sampling.⁴

Evaluating the performance of an image restoration method is perhaps a bit subtle. In Figure 15.4, the three posterior means look more like the true image than the raw data, but the raw data X seems visually almost as “clear.” One measure of performance might be the average mean square error, $\text{MSE} = \sum_z (\hat{\theta} - \theta)^2 / (nm)$. The raw data X is the UMVU estimator here and has $\text{MSE} = 9.0016$, very close to the expected MSE which is exactly 9, the common variance of the X_z . Mean squared errors for the posterior means are 0.9786, 1.0182, and 1.5127 for the undersmoothed, matched, and oversmoothed priors, respectively. Surprisingly, by this measure the image using the undersmoothed prior is a bit better than the image that matches the covariance between values at adjacent pixels.

⁴ Actually, taking advantage of the normal structure in this particular example, means for the Gibbs sampling Markov chain can be found recursively and converge to the posterior mean. This approach was used to produce the images in Figure 15.4, iterating the recursion numerically. See Problem 15.9.

15.7 Problems

1. Consider Example 15.3 in the balanced case where $n_1 = \cdots = n_I$. Derive a formula for

$$P[T = 1 | Y_{ij}, i = 1, \dots, I, j = 1, \dots, n_j].$$

2. Verify the relation (15.1).
3. Find the stationary distribution for a Markov chain on $\mathcal{X} = \mathbb{R}$ with kernel Q given by

$$Q_x = N(cx, 1), \quad x \in \mathbb{R},$$

where c is a fixed constant with $|c| < 1$.

4. Let Q be the transition kernel for a Markov chain on $\mathcal{X} = \{0, 1, 2, \dots\}$ given by

$$\mathbf{Q}_{ij} = \begin{cases} 1, & i = 0, j = 1; \\ 1/2, & i > 0, j = i + 1; \\ 1/2, & i > 0, j = 0; \\ 0, & \text{otherwise,} \end{cases}$$

where $\mathbf{Q}_{ij} \stackrel{\text{def}}{=} Q_i(\{j\})$. (Here $\mathbf{Q}$ might naturally be viewed as an infinite-dimensional transition matrix.) So at each stage, this chain has an equal chance of increasing by one or falling back to zero. The Markov chain with transition kernel Q has a unique stationary distribution π . Find the mass function for π .

5. Consider using the Metropolis–Hastings algorithm to sample from the standard normal distribution. Assume that the jump kernel J is given by

$$J_x = N(x/2, 1), \quad x \in \mathbb{R}.$$

Give a formula for r and find the chance the chain does not move when it is at position x ; that is, $P(X_1 = x | X_0 = x)$.

6. Consider using the Metropolis–Hastings algorithm to sample from a discrete distribution on $\mathcal{X} = \{1, \dots, 5\}$ with mass function

$$f(x) = cx, \quad x = 1, \dots, 5,$$

for some constant c . Suppose the transition matrix for the jump kernel J is

$$\mathbf{J} = \begin{pmatrix} 1/2 & 1/2 & 0 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 1/2 & 0 & 1/2 \\ 0 & 0 & 0 & 1/2 & 1/2 \end{pmatrix}.$$

Find the transition matrix $\mathbf{Q}$ for the Metropolis–Hastings chain. Check that the vector $\boldsymbol{\pi}$ corresponding to the mass function f is a left-eigenvector for $\mathbf{Q}$ with unit eigenvalue.

7. Consider Gibbs sampling with target distribution

$$\pi = N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} \right).$$

Find π_1 and π_2 if $X_0 = x$ and $Y_0 = y$, so that π_0 is a point mass at (x, y) .

8. Consider Gibb's sampling for an absolutely continuous distribution with density

$$f(x, y) = \begin{cases} ce^{-x-y-xy}, & x > 0, y > 0; \\ 0, & \text{otherwise,} \end{cases}$$

for some constant c . Find the joint density of X_1 and Y_1 if $X_0 = Y_0 = 1$.

9. Consider using Gibbs sampling for the posterior distribution of an image in the model considered in Section 15.6. Let $\Theta(n)$, $n \geq 0$, be images generated by Gibbs simulation, and let $\mu(n)$ denote the mean of $\Theta(n)$. Use smoothing and (15.12) to derive an equation expressing $\mu(n+1)$ as a function of $\mu(n)$. These means $\mu(n)$ converge to the true mean of the posterior distribution as $n \rightarrow \infty$, so the equation you derive can be used to find the posterior mean by numerical recursion.
10. Consider Bayesian image restoration for the model considered in Section 15.6 when the prior density has form

$$\lambda(\theta) \propto_{\theta} \exp \left[-\frac{a}{2} \sum_z \theta_z^2 + b \sum_{\{z_1, z_2\} \in \mathcal{E}_1} \theta_{z_1} \theta_{z_2} + c \sum_{\{z_1, z_2\} \in \mathcal{E}_2} \theta_{z_1} \theta_{z_2} \right],$$

where

$$\mathcal{E}_1 = \{(z_1, z_2) \in \mathcal{T}^2 : \|z_1 - z_2\| = 1\}$$

and

$$\mathcal{E}_2 = \{(z_1, z_2) \in \mathcal{T}^2 : \|z_1 - z_2\| = \sqrt{2}\}.$$

With this prior, it is natural to partition the pixels $\mathcal{T}$ into four sets, $\mathcal{T}_{00}$, $\mathcal{T}_{01}$, $\mathcal{T}_{10}$, and $\mathcal{T}_{11}$, given by

$$\mathcal{T}_{ab} = \{(i, j) \in \mathcal{T} : i \equiv a \pmod{2}, j \equiv b \pmod{2}\},$$

for $a = 0, 1$ and $b = 0, 1$. Describe how to implement Gibb's sampling from the posterior distribution in this case. As in (16.11), for $z \in \mathcal{T}_{00}$ find the conditional distribution of Θ_z given $\Theta_{\tilde{z}} = \theta_{\tilde{z}}$, $\tilde{z} \in \mathcal{T}_{01} \cup \mathcal{T}_{10} \cup \mathcal{T}_{11}$, and $X = x$.